Pointwise Frequency Attention and Coarse-to-Fine Learning With Image-to-Instance Level Supervision for Fine-Grained Object Detection in Remote Sensing Images

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***Abstract*—Fine-grained object detection (FGOD) models aim to distinguish different subcategories within the same coarse-grained category. Although two-stage object detection frameworks are extensively employed in current FGOD models due to their high detection accuracy, they still face two major problems. First, insufficient discriminability of the backbone features results in frequent misclassification among similar subcategory objects. Second, the limited capability of the backbone features to represent object direction leads to localization deviation. To address the first problem, a coarse-to-fine image-level supervised feature discriminability enhancement (C2FIm-FDE) module is proposed. Guided by the generated coarse- and fine-grained image-level category labels, the C2FIm-FDE module progressively strengthens the discriminability of backbone features at the image level. To tackle the second problem, a pointwise frequency attention based on the discrete wavelet transform (PFA-DWT) is proposed. By adaptively combining multiband frequency features and leveraging the dual localization capability of the DWT in the spatial and frequency domains, the PFA-DWT module achieves accurate fusion of frequency-domain features encoding local object orientation information with spatial features via a pointwise attention mechanism. To further enhance the discriminability of similar subcategories, a coarse-to-fine instance-level supervised detection (C2FIn) head is proposed, which gradually focuses on differences among subcategories within the same coarse-grained category by sequentially introducing coarse- and fine-grained instance-level labels. Ablation studies validate the effectiveness of the C2FIm-FDE, PFA-DWT modules, and C2FIn head, as well as their arbitrary combinations. Quantitative comparisons with popular FGOD models on the two large-scale public datasets, FAIR1M-1.0 and FAIR1M-2.0, demonstrate the excellent detection accuracy and generalization capability of our FGOD model. Source code is available at https://github.com/PFA-DWT/PFA-C2FI**

***Index Terms*—Coarse-to-fine image-level supervised feature discriminability enhancement (C2FIm-FDE) module, C2FIn head, fine-grained object detection (FGOD), pointwise frequency attention based on discrete wavelet transform (PFA-DWT) module, remote sensing images (RSIs).**

I. Introduction

TRADITIONAL object detection models in remote sensing images (RSIs) mainly identify coarse-grained categories [1], [2], such as airplane, ship, and vehicle, whereas fine-grained object detection (FGOD) models focus on identifying subcategories within a coarse category, e.g., C919, A321, and Boeing 737. The FGOD in RSIs has broad applications in Earth observation [3], urban monitoring [4], disaster management [5], and so on.

The current object detection models are roughly divided into three categories, i.e., anchor-based one-stage models [6], [7], [8], anchor-based two-stage models [9], [10], and anchor-free models [11]. Owing to the severe interclass similarity among subcategories, FGOD models are required to achieve high detection accuracy. Consequently, most existing models adopt anchor-based two-stage object detection frameworks, which offer superior detection performance. Although recent advances in deep learning have substantially improved FGOD capability [12], [13], [14], [15], [16], [17], [18], current models still face the following two challenges.

The first issue is the limited discriminability of backbone features, which leads to the misidentification of similar subcategories. To address the confusion among subcategories, most existing models focus on improving the architecture of the detection head or loss function [19], [20], [21], [22], whereas only a few studies explore improvements to the backbone network [23], [24]. Benefiting from the guidance of instance-level supervision, the improvement of the object detection head has achieved remarkable results. In contrast, due to the lack of direct supervisory guidance, existing studies of backbone networks usually depend on heuristic architectural modifications driven by empirical experience, resulting in limited improvements in feature discriminability. Since backbone features are jointly utilized by the region proposal network (RPN) and detection heads, this limitation is detrimental to improve proposal quality and alleviate misclassification among subcategories. As shown in Fig. 1(a), the baseline model

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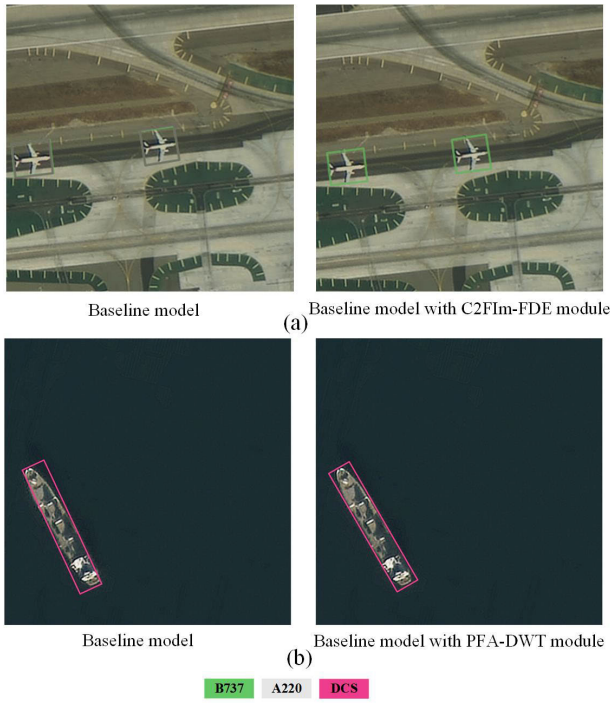


Fig. 1. Illustration of the research motivation of this article. (a) Misclassification of two airplanes induced by insufficient discriminability of backbone features. (b) Inaccurate localization of a ship induced by the limited capability of backbone features for object orientation. The above RSIs are selected from the FAIR1M-1.0 dataset [25].

with a traditional backbone mistakenly identifies two Boeing airplanes as A220.

The second issue arises from the limited capability of backbone features for representing object direction, which leads to inaccurate localization of oriented objects. Existing FGOD models generally adopt CNN-based backbones to extract spatial features; however, the horizontally aligned rectangular convolution kernels in CNNs are limited in representing objects with arbitrary orientations [26], [27]. In contrast, frequency-domain features contain rich direction information, which supports accurate representation of object direction [28], [29]. Most existing frequency-domain attention modules apply trigonometric transforms to spatial feature maps to obtain frequency-domain representations, followed by convolution over multifrequency features to generate channel attention for enhancing spatial features [30], [31], [32]. Nevertheless, trigonometric functions perform only global frequency-domain transformations and lack local spatial localization capability [33], [34], [35], which is why existing frequency-domain attention modules are generally restricted to channel attention. In other words, most frequency-domain attention modules fail to effectively capture local frequency-domain features of objects. As shown in Fig. 1(b), the baseline model with CNN-based backbone has a deviation from the angle positioning of a ship.

To address the first issue, a coarse-to-fine image-level supervised feature discriminability enhancement (C2FIm-FDE) module is proposed to improve the discriminability of backbone features. Considering the lack of direct supervisory guidance for the training of backbone features, coarse- and fine-grained image-level labels are inferred from instance-level labels and utilized to guide a coarse-to-fine progressive enhancement of the backbone features. As shown in Fig. 1(a), after integrating the C2FIm-FDE module, the baseline model is able to accurately identify all airplanes.

To address the second issue, a pointwise frequency attention based on the discrete wavelet transform (PFA-DWT) is proposed to improve the orientation-aware feature representation of the backbone network. By utilizing the spatial–frequency dual localization capability of DWT, the PFA-DWT module can capture both global structural information and local frequency-domain features, effectively compensating for the limitations of trigonometric-based frequency transforms. In addition, the PFA-DWT module can adaptively fuse multiband DWT features to generate pointwise attention with both spatial and channel awareness, enabling accurate integration of frequency-domain and spatial features. As shown in Fig. 1(b), the baseline model achieves the precise localization of the ship after incorporating the PFA-DWT module.

To further enhance the discriminability of similar subcategories, a coarse-to-fine instance-level supervised detection (C2FIn) head is proposed. Through a coarse-to-fine instance-level supervision strategy, the detection head is gradually encouraged to capture discriminative differences among subcategories.

The main contributions of this article are summarized as follows.

1) A C2FIm-FDE module is proposed and integrated into the backbone network to address the limited discriminability of backbone features. By leveraging generated image-level supervision, the C2FIm-FDE module guides a coarse-to-fine progressive enhancement of the discriminability of the backbone features.
2) A PFA-DWT module is proposed to tackle the insufficient orientation representation of the backbone features. Leveraging the spatial–frequency dual localization capability of the DWT, the PFA-DWT module adaptively integrates global and local frequency-domain features and generates pointwise attention to accurately fuse frequency-domain and spatial features.
3) A C2FIn head is proposed to further improve the discriminability of similar subcategories. Through the progressive integration of coarse- and fine-grained instance-level supervision, the C2FIn head is gradually guided to focus on the difference of subcategories.

II. Related Work

FGOD builds upon oriented object detection (OOD) to further identify similar subcategories. Accordingly, this section first reviews related works on OOD, then surveys representative studies on FGOD, and finally introduces the task-consistent focal (TF) loss [36] employed in the baseline model.

A. Oriented Object Detection

The current OOD models are roughly divided into three categories, i.e., anchor-based one-stage models, anchor-based two-stage models, and anchor-free models.

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Anchor-based one-stage OOD models predefine multiple anchors at each feature point and employ fully convolutional networks to simultaneously predict the class confidence score (CCS) and location offset of each anchor. To address the mismatch between predefined anchors and objects with extreme aspect ratios, S^2A-Net [37] proposes an alignment convolution to generate high-quality anchors. R^3Det [38] proposes a feature refinement module to more accurately extract features of objects with extreme aspect ratios and introduces an improved SkewIoU loss to achieve more precise prediction of object rotation angles. Other classical works include DAL [39], HRDet [40], and so on.

Two-stage OOD models first utilize an RPN to generate proposals from predefined anchors and then predict the CCSs and localization offsets for all proposals. RoI Transformer [41] first generates horizontal proposals from predefined anchors and then transforms them into oriented proposals through a region of interest (RoI) learner. Oriented R-CNN [42] proposes an oriented RPN, which directly generates oriented proposals based on predefined anchors, significantly improving computational efficiency while preserving proposal quality. AOPG [43] initially employs an anchor-free subnetwork to generate high-quality anchors, upon which high-quality oriented proposals are generated. Other representative works include Strip R-CNN [44], Info-FPN [45], and so on.

Anchor-free OOD models avoid predefined anchors and directly obtain oriented bounding boxes by predicting the distances from each feature point to the four sides of the box, together with the orientation angle and CCS. DRN [46] adaptively selects between traditional convolutional and strip convolutional features to handle irregular object representation and further introduces a dynamic convolution kernel generator in the detection head to extract instance-level features. Oriented RepPoints [47] represents each object as a set of adaptively adjustable points that flexibly evolve according to object geometry, guiding deformable convolution through dynamic point set adjustment for precise feature extraction of irregular objects. Other representative works include DFDet [11], R2IPoint [48], and so on.

B. Fine-Grained Object Detection

Most existing FGOD models focus on improving the feature attention modules and loss functions, which are introduced as follows.

Feature attention modules are still predominantly based on spatial or channel attention mechanisms. SFRNet [49] uses multihead attention and a local hashing algorithm to compute spatial and channel attention along the spatial and channel dimensions, respectively, and enhances features by combining the two attention mechanisms. PETDet [26] strengthens RPN feature representations by cascading channel and spatial attention, thereby producing higher-quality proposals. Furthermore, it leverages bilinear fusion of adjacent channels to more effectively exploit spatial details from low-level features for distinguishing similar subcategories. EIRNet [50] proposes an FGOD model specifically designed for ship targets. It first generates pixel-level ground-truth maps for head, tail, and global regions of the ship to supervise the prediction of corresponding spatial attention maps, which are then aggregated to form more accurate and complete spatial attention maps, thereby enhancing the discriminability of backbone features. To address the insufficient feature representation of fine-grained objects under scene-centric or back-view imaging conditions, CGL [51] proposes a concept-guided learning framework. It assumes that the feature of each fine-grained category can be decomposed into a shared component inherited from its corresponding coarse-grained category and a distinctive component that captures category-specific attributes. The distinctive component is further exploited to generate channel attention, thereby guiding the recognition of fine-grained categories. Other important related works include SPA [52], CMAM-FAM [53], P-CNN [54], and so on.

Existing FGOD models typically employ loss functions designed to maximize interclass differences. DRNet [55] introduces a confusion minimization loss that defines sample separability and assigns higher weights to samples with low separability, thereby mitigating confusion among similar subcategories. PCLDet [56] constructs a prototype feature for each subcategory and formulates a loss function by maximizing the similarity between the prototype and features of samples from the same class while minimizing the similarity to features from different categories. The prototype features are iteratively updated during training. The category of a testing image is determined by comparing its feature representation with the prototype features of all categories during inference. LOOD [57] leverages text prompts to enhance the capability of OOD models in recognizing fine-grained categories. Specifically, its core idea is to perform an orthogonal decomposition of RoI features, where the components that are uncorrelated with the textual embeddings of coarse-grained category labels are regarded as discriminative features for fine-grained categories. These discriminative features of positive samples are then stored in a dynamically updated queue. Finally, a contrastive loss is computed between the features in the queue and the positive samples within the current minibatch, aiming to minimize feature similarity among different fine-grained categories. Other representative works include BSNet [58], ISCL [59], and so on.

C. TF Loss

A TF loss [36], denoted as L_{TF}, is utilized in the baseline model to enhance the consistency between classification and regression and highlight hard samples. Its formulation is given as follows:

$$L_{TF} = TFC\left(cls,\ IoU_{ob},\ L^{C}\right) + TFR\left(cls,\ IoU_{ob},\ Loc_{ob},\ L^{Loc}\right) \tag{1}$$

$$TFC\left(cls, IoU_{ob}, L^{C}\right) = \begin{cases} e^{\beta_1(1-IoU_{ob})} \times CE\left(cls, L^{C}\right), & \text{if } p \in S_P \\ FL\left(cls, L^{C}\right), & \text{if } p \in S_N \end{cases} \tag{2}$$

$$TFR\left(cls, IoU_{ob}, Loc_{ob}, L^{Loc}\right) = e^{\beta_2|IoU_{ob}-cls|} \times SL\left(Loc_{ob}, L^{Loc}\right) \tag{3}$$

where $TFC(\cdot)$ denotes TF classification (TFC) loss, $TFR(\cdot)$ denotes TF regression (TFR) loss, p denotes the proposal

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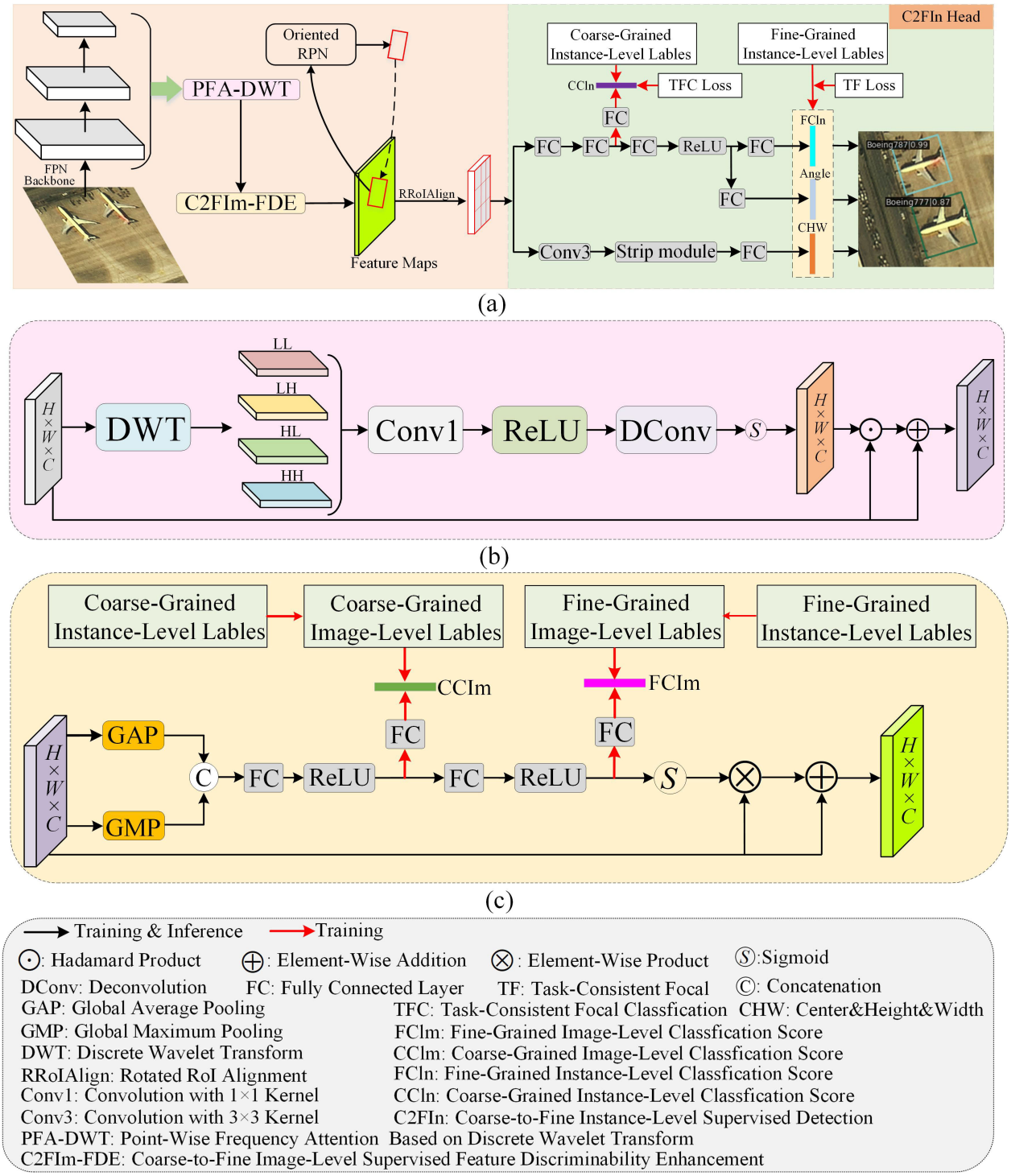


Fig. 2. Illustration of the proposed method. (a) Overall framework and the details of C2FIn head. (b) and (c) Framework of PFA-DWT and C2FIm-FDE modules.

generated by oriented RPN, S_P and S_N denote the sets of positive and negative samples, respectively, $CE(\cdot)$ denotes cross-entropy loss [60], cls denotes the CCS of each proposal, $SL(\cdot)$ denotes smooth L1 loss [60], $FL(\cdot)$ denotes focal loss [61], IoU_{ob} denotes the Intersection over Union (IoU) between the predicted oriented bounding box and the corresponding ground-truth box, $Loc_{ob} \in \mathbb{R}^5$ denotes the predicted object location, $L^{Loc} \in \mathbb{R}^5$ denotes the localization labels, L^C denotes instance-level category labels, and β_1 and β_2 are hyperparameters in $TFC(\cdot)$ and $TFR(\cdot)$, respectively.

III. Proposed Method

A. Overview

The overall framework of the proposed method is illustrated in Fig. 2(a). First, multilevel features produced by the feature pyramid network (FPN) are sent to the PFA-DWT module,

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where spatial and frequency-domain features are fused in a pointwise manner. The fused features are then processed by the C2FIm-FDE module to enhance the discriminability of backbone features. Subsequently, oriented proposals are generated by the oriented RPN [42], and proposal-level features are extracted through a rotated RoI alignment (RRoIAlign) operation. Finally, the proposal features are sent to the C2FIn head for predicting CCS and localization offsets of each proposal, and the last detection results are obtained after nonmaximum suppression (NMS).

B. Pointwise Frequency Attention Based on Discrete Wavelet Transform

The overall architecture of the PFA-DWT module is illustrated in Fig. 2(b). First, the multilevel spatial features extracted by the FPN backbone are decomposed via DWT

$$(F_{LL}, F_{LH}, F_{HL}, F_{HH}) = DWT(F_S) \tag{4}$$

where $F_S \in \mathbb{R}^{H\times W\times C}$ denotes the spatial features of each level, H, W, and C denote the height, width, and number of channels of the F_S, respectively, $DWT(\cdot)$ denotes discrete wavelet transform, and F_{LL}, F_{LH}, F_{HL}, and $F_{HH} \in \mathbb{R}^{H/2\times W/2\times C}$ denote the low-frequency (LL), horizontal high-frequency (LH), vertical high-frequency (HL), and diagonal high-frequency (HH) components, respectively.

Second, the above four frequency-domain components are adaptively fused according to the following equation:

$$F_{AF} = ReLU(Conv1(CAT(F_{LL}, F_{LH}, F_{HL}, F_{HH}))) \tag{5}$$

where $F_{AF} \in \mathbb{R}^{H/2\times W/2\times C}$ denotes the fused features, $CAT(\cdot)$ denotes channelwise concatenation, $Conv1(\cdot)$ denotes a convolution operation with 1×1 kernel, and $ReLU(\cdot)$ denotes the ReLU activation function [62].

Subsequently, a PFA map is generated from F_{AF}, denoted as $M_{PFA} \in \mathbb{R}^{H\times W\times C}$, which can be formulated as follows:

$$M_{PFA} = Sig(DConv(F_{AF})) \tag{6}$$

where $DConv(\cdot)$ denotes a deconvolution operation and $Sig(\cdot)$ denotes sigmoid activation function.

Finally, pointwise fusion of spatial and frequency-domain information is performed according to the following equation:

$$F_{SF} = (F_S \odot M_{PFA}) \oplus F_S \tag{7}$$

where $F_{SF} \in \mathbb{R}^{H\times W\times C}$ denotes the features obtained by fusing spatial and frequency-domain information, $\odot$ denotes the Hadamard product, and $\oplus$ denotes the elementwise addition.

In summary, the PFA-DWT module fully exploits the spatial–frequency dual localization capability of the DWT. Frequency-domain transformation is utilized to capture orientation information, compensating for the limitations of spatial features, while PFA enables accurate fusion of spatial and frequency-domain information.

C. Coarse-to-Fine Image-Level Supervised Feature Discriminability Enhancement Module

The overall architecture of C2FIm-FDE module is illustrated in Fig. 2(c). First, the C2FIm-FDE module takes the output feature F_{SF} of the PFA-DWT module as input, and the corresponding coarse-grained image-level features, denoted as $F_{CIm} \in \mathbb{R}^{1\times 1\times C}$, are obtained according to the following equation:

$$F_{CIm} = ReLU(FC(CAT(GAP(F_{SF}), GMP(F_{SF})))) \tag{8}$$

where $GAP(\cdot)$ and $GMP(\cdot)$ denote global average pooling and global max pooling, respectively, and $FC(\cdot)$ denotes a fully connected convolution operation.

Next, the fine-grained image-level features, denoted as $F_{FIm} \in \mathbb{R}^{1\times 1\times C}$, are obtained according to the following equation:

$$F_{FIm} = ReLU(FC(F_{CIm})). \tag{9}$$

Finally, the enhanced backbone features, denoted as $F_{EIm} \in \mathbb{R}^{H\times W\times C}$, are obtained by using the following equation:

$$F_{EIm} = (F_{SF} \otimes Sig(F_{FIm})) \oplus F_{SF} \tag{10}$$

where $\otimes$ denotes the elementwise product.

It is worth noting that the generation of F_{CIm} and F_{FIm} is guided by the coarse- and fine-grained image-level labels. Accordingly, the loss function of the C2FIm-FDE module, denoted as L_{CF}, is formulated as follows:

$$\begin{aligned} L_{CF} &= BCE(CCIm, LIm^c) + BCE\left(FCIm, LIm^f\right) \\ &= BCE(FC(F_{CIm}), LIm^c) + BCE\left(FC(F_{FIm}), LIm^f\right) \end{aligned} \tag{11}$$

where $LIm^c \in \mathbb{R}^{NC}$ denotes the coarse-grained image-level labels, NC denotes the number of coarse-grained categories, $CCIm$ and $FCIm$ denote the CCSs of the coarse- and fine-grained categories, respectively, and $BCE(\cdot)$ denotes the binary cross-entropy loss. Specifically, $LIm^c \in \mathbb{R}^{NC}$ can be derived from coarse-grained instance-level labels, i.e., $Lim^c(i) = 1$ if the input image contains at least an instance belonging to the ith coarse-grained category, where $i \in [1, NC]$; otherwise, $Lim^c(i) = 0$. Similarly, $LIm^f \in \mathbb{R}^{NF}$ denotes the fine-grained image-level labels and is obtained from fine-grained instance-level labels following the same procedure as LIm^c, where NF denotes the number of fine-grained categories.

In summary, the C2FIm-FDE module generates fine-grained image-level features under the guidance of both coarse- and fine-grained image-level labels and leverages them to enhance the backbone features. Consequently, the enhanced backbone features achieve stronger discriminability at the image-level classification stage, laying a solid foundation for subsequent instance-level classification.

D. Coarse-to-Fine Instance-Level Supervised Detection Head

The overall architecture of C2FIn head is illustrated by the light green region in Fig. 2(a). First, the coarse-grained category features of each proposal, denoted as $F_{CIn} \in \mathbb{R}^N$, are obtained through the following equation:

$$F_{CIn} = FC\left(FC\left(F_p\right)\right) \tag{12}$$

where $F_p \in \mathbb{R}^{7\times 7\times C}$ denotes the pooled feature of an oriented proposal and N denotes the number of dimensions of the F_{CIn}, Subsequently, the fine-grained category features, denoted as

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$F_{FIn} \in \mathbb{R}^N$, are further obtained according to the following equation:

$$F_{FIn} = ReLU\left(FC\left(F_{CIn}\right)\right). \tag{13}$$

Inspired by Strip R-CNN [44], the F_{FIn} is used to predict both the fine-grained CCSs and angle offset of each oriented proposal, denoted as $FCIn \in \mathbb{R}^{NC}$ and θ_o, respectively.

Subsequently, the offsets of the coordinates of the center point, width, and height of each oriented proposal, denoted as (x_o, y_o), w_o, and h_o, respectively, are obtained through the following equation:

$$(x_o, y_o, w_o, h_o) = FC\left(Strip\left(Conv3\left(F_p\right)\right)\right) \tag{14}$$

where $Conv3(\cdot)$ denotes a convolution operation with 3×3 kernel and $Strip(\cdot)$ denotes the convolution operations introduced in strip module [44]. Finally, the predicted object location Loc_{ob} is obtained through the following equation:

$$\begin{aligned} Loc_{ob} &= Loc_p + (x_o, y_o, w_o, h_o, \theta_o) \\ &= Loc_p + (x_o, y_o, w_o, h_o, FC\left(F_{FIn}\right)) \end{aligned} \tag{15}$$

where $Loc_p \in \mathbb{R}^5$ denotes the location of oriented proposal predicted by the oriented RPN [42].

To enhance the consistency between classification and regression tasks, the predictions of F_{CIn} and Loc_{ob} in the above procedure are jointly supervised by instance-level category and localization labels. Accordingly, the loss functions of C2FIn head, denoted as L_{CH}, are defined as follows:

$$L_{CH} = L_{CC} + L_{TF} \tag{16}$$

$$\begin{aligned} L_{CC} &= TFC\left(CCIn, IoU_{ob}, LIn^c\right) \\ &= TFC\left(FC\left(F_{CIn}\right), IoU_{ob}, LIn^c\right) \end{aligned} \tag{17}$$

$$\begin{aligned} L_{TF} &= TFC\left(FCIn, IoU_{ob}, LIn^f\right) \\ &\quad + TFR\left(FCIn, IoU_{ob}, Loc_{ob}, L^{Loc}\right) \\ &= TFC\left(FC\left(F_{FIn}\right), IoU_{ob}, LIn^f\right) \\ &\quad + TFR\left(FC\left(F_{FIn}\right), IoU_{ob}, Loc_{ob}, L^{Loc}\right) \end{aligned} \tag{18}$$

$$IoU_{ob} = IoU\left(Loc_{ob}, L^{Loc}\right) \tag{19}$$

where $CCIn \in \mathbb{R}^{NC}$ denotes the coarse-grained CCSs of each proposal, $LIn^c \in \mathbb{R}^{NC}$ and $LIn^f \in \mathbb{R}^{NF}$ denote the coarse- and fine-grained instance-level category labels of each proposal, respectively, L_{CC} denotes the coarse-grained instance-level classification loss, and L_{TF}, $TFC(\cdot)$, and $TFR(\cdot)$ have been introduced in Section II-C, respectively.

The total loss of our FGOD model, denoted as L, is defined as

$$L = L_{RPN} + L_{CF} + L_{CH} \tag{20}$$

where L_{RPN} denotes the loss of oriented RPN, and its details can be found in Oriented R-CNN [42].

IV. Experiments

A. Experiment Setup

1) Datasets: Experiments are conducted on two large-scale public benchmarks for the evaluation of FGOD performance, i.e., FAIR1M-1.0 [25] and FAIR1M-2.0 datasets. FAIR1M-1.0 contains 24 625 RSIs with spatial resolutions ranging from 1000 × 1000 to 10 000 × 10 000 pixels, among which 16 488 RSIs are used for training, and the remaining RSIs are used for testing. The dataset includes five coarse categories, i.e., airplane, ship, vehicle, court, and road. These categories are further divided into 37 subcategories, i.e., Boeing 737 (B737), Boeing777 (B777), Boeing 747 (B747), Boeing 787 (B787), AirbusA321 (A321), Airbus A220 (A220), Airbus A330 (A330), Airbus A350 (A350), COMAC C919 (C919), COMAC ARJ21 (ARJ21), other-airplane (OA), passenger ship (PS), motorboat (MB), fishing boat (FB), tugboat (TB), engineering ship (ES), liquid cargo ship (LCS), dry cargo ship (DCS), warship (WS), other-ship (OS), small car (SC), bus (BUS), cargo truck (CT), dump truck (DT), van, trailer (TRI), tractor (TRC), truck tractor (TT), excavator (EX), other-vehicle (OV), baseball field (BF), basketball court (BC), football field (FF), tennis court (TC), roundabout (RA), intersection (IS), and bridge (BR). FAIR1M-2.0 consists of 42 796 RSIs, with 16 488 RSIs for training, 8287 for validation, and 18 021 for testing. Both the training and validation subsets are used for training. To the best of our knowledge, FAIR1M-2.0 is currently the largest publicly available dataset for FGOD. The object categories in FAIR1M-2.0 are identical to those in FAIR1M-1.0. Note that all experimental results on the above two datasets are evaluated on the official evaluation server,[1] and the official evaluation results of both datasets exclude three subcategories, i.e., OA, OS, and OV.

2) Implementation Details: Oriented R-CNN [42] combined with the TF loss [36] is adopted as the baseline model, with ResNet50-FPN [63] serving as the backbone network. Our final trained FGOD model is denoted as PFA-C2FI, which is optimized using stochastic gradient descent (SGD), where the momentum and weight decay are set to 0.9 and 0.0001, respectively. The training batch size, learning rate, and IoU threshold in NMS are consistent with the Oriented R-CNN. The number of training epochs is set to 12. All experiments are conducted on the MMRotate[2] platform based on the PyTorch framework, and the source code is run on four NVIDIA TITAN RTX GPUs (24 GB ×4 memory). The hyperparameters β_1 in (2) and β_2 in (3) follow the same settings as those in the TF loss [36]. To quantitatively evaluate the performance of the proposed PFA-C2FI model, the average precision (AP) for each category and the mean AP (mAP) over all categories are adopted.

B. Ablation Study

In this section, overall ablation studies are first conducted to assess the effectiveness of the PFA-DWT module, the C2FIm-FDE module, and the C2FIn head. Then, more detailed ablation studies are performed to analyze within each component.

1) Overall Ablation Study: As shown in Table I, when the PFA-DWT module, the C2FIm-FDE module, and the C2FIn head are individually incorporated into the baseline model, the mAP on FAIR1M-1.0 is improved by 1.98%, 2.46%, and 1.84%, respectively, and 2.19%, 2.56%, and 1.46% on FAIR1M-2.0, demonstrating the effectiveness of

[1] https://gaofen-challenge.com/

[2] https://github.com/open-mmlab/mmrotate/

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TABLE I

ABLATION STUDIES OF THE PFA-DWT MODULE, THE C2FIM-FDE MODULE, AND THE C2FIN HEAD ON THE FAIR1M-1.0 AND FAIR1M-2.0 DATASETS

PFA-DWT	C2FIm-FDE	C2FIn Head	mAP (%)	
			FAIR1M-1.0	FAIR1M-2.0
×	×	×	40.55 (Baseline)	45.01 (Baseline)
√	×	×	42.53 (+1.98)	47.20 (+2.19)
×	√	×	43.01 (+2.46)	47.57 (+2.56)
×	×	√	42.39 (+1.84)	46.47 (+1.46)
√	√	×	44.00 (+3.45)	48.52 (+3.51)
×	√	√	43.89 (+3.34)	48.36 (+3.35)
√	×	√	43.82 (+3.27)	48.31 (+3.30)
√	√	√	44.36 (+3.81, Ours)	48.67 (+3.66, Ours)

TABLE II

ABLATION STUDY FOR THE WAVELET DECOMPOSITION SETTINGS OF PFA-DWT MODULE ON THE FAIR1M-1.0 DATASET

LL	LH	HL	HH	mAP (%)
×	×	×	×	40.55 (Baseline)
×	√	√	√	38.66 (-1.89)
√	×	×	×	41.31 (+0.76)
√	√	×	×	41.74 (+1.19)
√	×	√	×	41.80 (+1.25)
√	×	×	√	41.63 (+1.08)
√	√	√	√	42.53 (+1.98)

each component. When the PFA-DWT module is combined with the C2FIm-FDE module and the C2FIn head, the mAP is improved by 3.45% and 3.27% on FAIR1M-1.0, and 3.51% and 3.30% on FAIR1M-2.0, respectively. When the C2FIm-FDE module is combined with the C2FIn head, the mAP improves by 3.34% and 3.35% on FAIR1M-1.0 and FAIR1M-2.0, respectively. When all three components are jointly integrated, the mAP achieves the highest improvements of 3.81% and 3.66% on FAIR1M-1.0 and FAIR1M-2.0, respectively. These results demonstrate the effectiveness of each component and the complementary benefits of their combinations.

2) Ablation Study for PFA-DWT Module: To further validate the effectiveness of the PFA-DWT module, detailed ablation studies are conducted under different wavelet decomposition settings. As shown in Table II, using only high-frequency subbands leads to worse performance than the baseline module, indicating that the low-frequency subband containing the main structural information is indispensable. The best performance is achieved when all four subbands are utilized, demonstrating that each subband contributes to the final detection results.

3) Ablation Study for C2FIm-FDE Module and C2FIn Head: To further validate the effectiveness of the C2FIm-FDE module and the C2FIn head, detailed ablation studies are conducted under different supervision settings. As shown in Table III, in the C2FIm-FDE module, the coarse-to-fine image-level supervision achieves better performance than using either coarse- or fine-grained supervision alone. In the C2FIn head, the coarse-to-fine instance-level supervision outperforms the use of fine-grained instance-level supervision alone. The best performance is achieved when both the C2FIm-FDE module and the C2FIn head adopt the coarse-to-fine supervision strategy.

TABLE III

ABLATION STUDY FOR C2FIM-FDE MODULE AND C2FIN HEAD ON THE FAIR1M-1.0 DATASET

C2FIm-FDE		C2FIn Head		mAP (%)
Coarse Supervision	Fine Supervision	Coarse Supervision	Fine Supervision	
×	×	×	√	40.55 (Baseline)
√	×	×	√	42.08 (+1.53)
×	√	×	√	42.55 (+2.00)
√	√	×	√	43.01 (+2.46)
√	×	√	√	43.16 (+2.61)
×	√	√	√	43.54 (+2.99)
√	√	√	√	43.89 (+3.34)

TABLE IV

COMPARISONS BETWEEN DWT AND OTHER FREQUENCY TRANSFORM METHODS FOR PFA-DWT MODULE ON THE FAIR1M-1.0 DATASET

FFT	DCT	DWT	mAP (%)
×	×	×	40.55 (Baseline)
√	×	×	41.40 (+0.85)
×	√	×	42.12 (+1.57)
×	×	√	42.53 (+1.98)

TABLE V

QUANTITATIVE COMPARISONS BETWEEN OUR MODELS WITH DIFFERENT BACKBONE NETWORKS ON THE FAIR1M-1.0 DATASET

Backbone	ResNet18 [63]	ResNet50 [63]	ResNet101 [63]	ARC-R50 [64]	LSKNet-S [65]	StripNet-S [44]
mAP (%)	42.68	44.36	44.65	44.97	45.07	45.74

C. Comparisons Between DWT and Other Frequency Transform Methods for PFA-DWT Module

To further validate the rationality of adopting DWT in the PFA-DWT module, the frequency transformation is replaced with fast Fourier transform (FFT) and discrete cosine transform (DCT), respectively, and quantitative comparisons are conducted. As shown in Table IV, the best performance is achieved when DWT is used, demonstrating that DWT, with

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TABLE VI

QUANTITATIVE COMPARISON ON THE FAIR1M-1.0 DATASET. ALL MODELS USE RESNET50-FPN AS THE BACKBONE AND ARE TRAINED FOR 12 EPOCHS. * DENOTES THE RESULTS REPRODUCED BY OURS. THE BOLDED NUMBERS DENOTE THE BEST RESULTS

Coarse category	Fine-grained category	RetinaNet [61]	R^3Det [38]	S^2ANet [37]	GLVE [66]	FRCNN [60]	DLL [67]	RoITrans [41]	ORCNN [42]	PCLDet [56]	PETDet [26]	SFRNet [49]	DRNet [55]	SRCNN* [44]	MSC-TF [36]	PFA-C2FI (Ours)
Airplane	B737	35.34	33.70	36.06	36.50	33.94	33.43	39.15	35.17	35.96	41.05	39.70	39.05	41.17	**43.41**	43.05
	B747	74.90	79.82	84.33	81.88	84.25	82.65	84.72	85.17	85.50	82.23	84.44	84.91	85.81	**86.68**	86.57
	B777	10.54	15.03	15.62	14.06	16.38	11.22	14.82	14.57	15.15	22.04	17.79	16.42	21.37	20.89	**23.15**
	B787	38.49	30.84	42.34	44.60	47.61	49.77	48.88	47.68	47.97	51.21	48.96	49.13	49.99	52.18	**52.76**
	C919	0.96	1.08	1.95	10.16	14.44	12.03	19.49	11.68	16.97	25.66	21.18	**52.50**	24.27	20.38	22.45
	A220	41.39	43.95	44.09	46.43	47.40	46.35	50.31	39.05	45.64	51.83	48.38	**68.26**	53.58	52.96	52.80
	A321	63.51	63.82	68.00	65.47	68.82	68.48	70.16	39.05	69.14	69.53	**71.25**	66.98	69.77	70.45	69.09
	A330	46.28	51.11	63.84	67.73	**72.71**	67.96	70.34	68.60	71.54	70.86	72.06	39.05	65.57	68.84	71.84
	A350	63.42	66.80	70.00	68.31	**76.53**	71.14	72.19	70.21	71.50	73.70	74.16	70.27	72.57	71.87	74.06
	ARJ21	2.29	3.76	12.10	25.97	26.59	24.29	33.72	25.32	38.68	36.90	31.47	**45.45**	33.61	31.72	34.34
Ship	PS	5.78	7.02	8.82	10.01	11.03	14.78	12.62	13.77	17.43	13.29	17.44	16.74	19.68	19.38	**21.01**
	MB	23.00	33.31	48.03	51.63	51.22	58.12	55.98	60.42	56.85	61.94	60.61	58.30	62.21	62.11	**63.31**
	FB	2.52	6.22	6.79	5.26	6.41	9.40	6.12	9.10	7.88	8.90	8.50	8.08	12.44	12.69	**12.90**
	TB	24.87	27.34	34.01	34.00	34.19	35.42	35.31	36.83	36.70	37.88	34.87	32.52	34.97	39.23	**39.55**
	ES	6.98	8.48	7.49	10.34	9.41	12.56	9.27	11.32	10.97	10.93	12.55	11.97	13.76	11.62	**15.41**
	LCS	7.39	17.90	18.30	14.53	15.17	21.36	15.95	21.83	19.91	22.05	19.71	22.07	24.68	23.76	**24.96**
	DCS	21.75	36.08	37.62	33.10	32.26	38.01	34.15	38.22	40.05	36.82	38.25	37.74	39.20	41.10	**41.31**
	WS	2.75	14.29	22.96	12.45	11.27	24.15	15.29	22.67	23.80	23.98	22.00	24.79	34.16	35.16	**35.16**
Vehicle	SC	37.22	50.72	61.57	54.39	54.56	55.66	57.55	57.62	56.78	**68.81**	57.73	58.39	60.99	62.35	61.96
	Bus	4.25	4.89	11.76	28.63	22.94	25.60	26.43	24.40	**35.10**	18.33	32.37	35.51	32.65	28.00	33.83
	CT	18.38	27.76	34.28	36.90	37.74	38.92	39.38	40.84	42.19	42.17	41.01	42.90	44.36	46.40	**46.46**
	DT	12.59	22.99	36.03	42.16	41.69	44.27	44.95	45.20	45.83	47.15	46.69	47.23	50.32	51.04	**51.24**
	VAN	26.44	41.70	54.62	48.52	48.23	50.59	53.69	54.01	52.01	**65.48**	54.08	53.59	56.68	57.47	57.55
	TRI	0.01	8.34	3.47	13.38	12.46	10.66	10.95	15.46	16.34	11.93	15.75	15.80	15.75	**21.50**	16.32
	TRC	0.02	1.42	0.96	1.39	2.44	4.46	2.13	2.37	6.43	2.09	**7.09**	4.26	4.42	5.24	5.96
	EX	0.13	6.00	7.24	12.19	11.35	16.35	10.99	13.55	18.08	8.51	15.97	17.42	19.16	19.49	**20.00**
	TT	0.03	0.18	0.40	0.19	0.32	0.28	0.60	0.24	0.61	0.41	0.37	1.05	1.99	2.12	**4.71**
Court	BC	27.51	40.39	38.44	45.83	45.18	46.90	46.93	48.18	48.45	42.64	49.43	50.43	57.21	57.81	**58.25**
	TC	79.20	77.78	80.44	77.99	77.75	81.21	79.29	78.45	78.13	78.60	79.17	82.25	**83.08**	82.20	81.40
	FF	55.44	53.28	56.34	59.27	52.05	46.03	56.72	60.79	62.55	63.51	59.90	60.53	**69.04**	68.84	68.41
	BF	87.89	87.85	87.47	86.08	87.19	87.55	87.21	88.43	88.23	87.40	87.90	86.96	89.97	**90.13**	89.42
Road	IS	54.38	53.73	50.76	58.19	58.71	57.66	58.21	57.90	59.01	57.48	59.48	58.51	60.98	61.61	**61.76**
	RA	23.91	21.72	16.67	19.45	19.38	16.76	21.98	17.57	20.21	20.82	22.05	24.29	25.80	24.75	**26.28**
	BR	4.38	17.02	17.24	22.82	20.76	24.39	23.31	28.63	31.67	22.57	32.76	26.58	37.42	39.75	**40.07**
mAP		26.58	31.07	34.71	36.47	36.83	38.26	38.49	38.85	40.39	40.55	40.74	40.87	43.20	43.63	**44.36**

TABLE VII

QUANTITATIVE COMPARISON ON THE FAIR1M-2.0 DATASET. ALL MODELS USE RESNET50-FPN AS THE BACKBONE AND ARE TRAINED FOR 12 EPOCHS. * DENOTES THE RESULTS REPRODUCED BY OURS. THE BOLDED NUMBERS DENOTE THE BEST RESULTS

Coarse category	Fine-grained category	RetinaNet [61]	R^3Det [38]	FRCNN [60]	GLVE [66]	S^2ANet [37]	RoITrans [41]	ORCNN [42]	PCLDet [56]	SFRNet [49]	PETDet [26]	DRNet [55]	SRCNN* [44]	PFA-C2FI(Ours)
Airplane	B737	37.72	39.09	42.84	42.95	41.90	47.76	44.24	42.77	45.29	**48.96**	43.28	45.38	47.27
	B747	89.45	91.19	93.28	92.98	92.21	94.13	93.27	93.44	92.97	94.57	93.68	94.63	**94.88**
	B777	30.17	32.44	35.63	37.59	34.73	38.62	36.81	36.96	40.87	43.24	39.87	39.25	**50.80**
	B787	50.07	55.25	62.19	57.19	54.04	62.21	59.27	61.60	61.21	**64.50**	63.72	64.24	62.91
	C919	0.64	0.66	3.58	6.31	0.66	12.18	6.35	12.79	15.96	**26.89**	16.49	13.47	21.26
	A220	53.37	52.24	53.51	53.02	55.05	54.96	55.29	52.53	54.17	**57.45**	54.20	55.91	56.82
	A321	67.00	68.25	68.28	64.11	71.02	69.07	70.25	69.76	69.36	**74.41**	68.07	71.49	71.43
	A330	41.19	45.28	58.65	57.95	55.66	60.11	61.05	64.11	61.50	62.84	59.82	63.03	**65.78**
	A350	54.88	65.42	63.46	65.02	63.72	66.17	68.04	68.18	68.26	68.45	70.25	73.45	**75.24**
	ARJ21	3.22	3.62	10.80	12.35	9.63	15.34	10.40	13.52	15.68	11.99	**18.92**	13.41	18.19
Ship	PS	5.67	10.98	10.80	12.75	11.92	13.51	14.12	14.93	14.45	14.13	17.00	15.15	**17.17**
	MB	29.80	40.11	51.82	56.51	51.11	57.76	61.61	58.04	61.47	**62.88**	62.03	62.08	60.79
	FB	6.25	9.79	12.65	15.99	12.67	20.89	26.14	25.12	26.58	24.61	29.38	30.60	**30.70**
	TB	23.11	28.32	28.35	27.63	29.03	30.78	30.55	30.34	30.56	31.59	33.01	31.13	**33.52**
	ES	12.21	12.96	11.45	13.23	13.76	15.12	15.97	17.03	**17.89**	16.53	18.39	17.83	17.16
	LCS	23.42	47.92	32.95	40.72	49.53	45.49	49.45	50.76	48.74	49.88	**50.85**	49.92	50.09
	DCS	31.23	46.39	35.34	42.42	50.07	49.50	51.95	52.11	50.79	49.46	**52.11**	51.57	52.01
	WS	8.70	25.79	13.88	20.95	27.55	29.06	33.01	30.22	31.98	28.92	36.78	**38.25**	37.46
Vehicle	SC	41.12	53.98	52.23	46.62	63.65	58.68	56.41	56.94	56.77	**70.17**	58.57	59.93	60.14
	Bus	10.05	5.76	24.12	26.47	14.71	30.25	31.81	34.10	32.58	31.31	**44.71**	36.00	34.90
	CT	26.79	37.74	45.51	44.21	43.63	50.35	51.29	49.85	50.93	52.91	53.10	54.99	**55.86**
	DT	20.90	31.41	44.54	41.74	41.41	48.80	48.74	49.71	49.00	51.14	50.36	**54.11**	53.99
	VAN	33.03	45.77	47.00	40.20	57.90	52.62	52.87	51.24	52.77	**68.30**	52.87	54.60	54.38
	TRI	0.77	10.92	8.63	12.95	8.23	13.18	15.89	15.81	17.56	12.94	16.54	16.66	**16.84**
	TRC	0.51	0.79	1.12	0.85	0.65	2.73	1.55	2.46	**2.47**	0.47	1.83	2.28	2.12
	EX	1.35	9.01	8.43	12.12	8.28	13.79	15.96	17.52	16.20	13.34	**20.85**	20.22	19.66
	TT	0.26	1.89	5.93	6.24	2.63	18.57	8.06	7.60	29.35	9.96	27.34	**40.19**	36.66
Court	BC	40.40	48.91	55.31	55.08	48.19	54.71	59.77	60.65	60.97	53.89	**62.89**	62.47	62.66
	TC	85.04	86.65	85.76	86.70	86.81	86.40	88.11	87.45	89.35	87.76	90.47	**90.78**	90.20
	FF	62.49	61.15	60.48	58.29	62.65	63.02	63.70	64.83	61.49	66.38	66.30	65.09	**68.58**
	BF	91.42	89.57	91.11	89.96	91.02	90.03	91.63	92.02	90.56	91.29	91.01	90.30	**93.10**
Road	IS	62.31	61.76	63.04	64.34	60.55	65.17	64.37	65.85	65.11	64.67	65.62	68.93	**69.14**
	RA	33.66	33.95	29.98	32.30	33.70	33.78	31.52	30.72	**35.13**	33.06	34.13	28.87	31.92
	BR	27.20	26.12	25.90	30.62	29.21	34.07	36.47	37.44	35.05	28.57	34.87	37.07	**41.01**
mAP		32.51	37.68	39.37	40.33	40.51	44.11	44.29	44.66	45.68	46.10	47.04	47.44	**48.67**

its joint spatial–frequency localization capability, is a more suitable choice.

D. Generalization Evaluation for Backbone

To evaluate the generalization capability of the proposed model, quantitative comparisons are conducted using different backbone networks. As shown in Table V, when stronger backbones than ResNet50 are adopted, the detection performance is further improved, demonstrating that the proposed model has strong generalization capability and performance scalability.

E. Comparisons With Popular FGOD Models

1) FAIR1M-1.0 Dataset: We conduct quantitative comparisons between the proposed PFA-C2FI model and 14 popular FGOD models on the FAIR1M-1.0 dataset. As shown in Table VI, the PFA-C2FI model achieves the highest mAP of 44.36%, which surpasses RetinaNet [61], R^3Det [38], S^2ANet [37], GLVE [66], FRCNN [60], DLL [67], RoITrans [41], ORCNN [42], PCLDet [56], PETDet [26], SFRNet [49], DRNet [55], SRCNN [44], and MSC-TF [36] by 17.78%, 13.29%, 9.65%, 7.89%, 7.53%, 6.10%, 5.87%, 5.51%, 3.97%, 3.81%, 3.62%, 1.16%, and 0.73%, respectively. These results verify the overall effectiveness of the PFA-C2FI model on the FAIR1M-1.0 dataset. Furthermore, our model achieves the highest AP in 18 out of 34 subcategories, demonstrating strong generalization capability.

2) FAIR1M-2.0 Dataset: The proposed PFA-C2FI model is quantitatively compared with 12 popular FGOD models on

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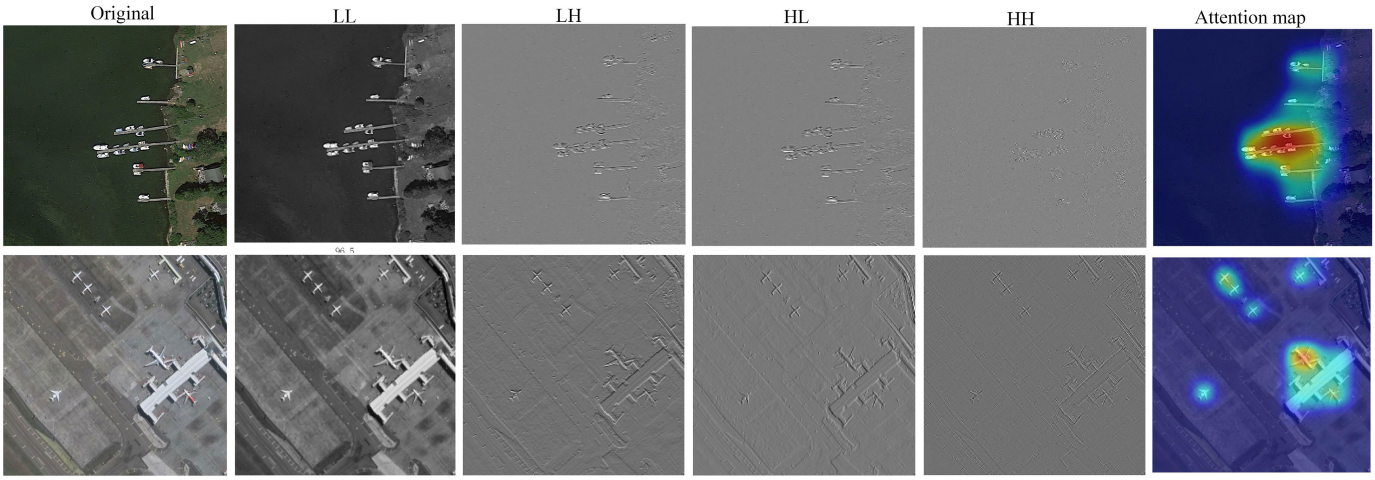


Fig. 3. Visualization of attention maps and frequency responses.

TABLE VIII
COMPARISON OF COMPUTATIONAL COST ON THE FAIR1M-1.0 DATASET

Method	GFLOPs	Parameters (M)	FPS	mAP (%)
FRCNN [63]	211.34	41.14	10.94	36.83
RoI Trans [41]	225.42	55.26	10.27	38.49
ORCNN [42]	211.45	41.16	10.93	38.85
MSC-TF [36]	216.75	41.36	10.90	43.63
Ours	217.19	49.33	10.88	44.36

the larger FAIR1M-2.0 dataset. As shown in Table VII, our model achieves the highest mAP of 48.67%, outperforming RetinaNet [61], R^3Det [38], FRCNN [60], GLVE [66], S^2ANet [37], RoITrans [41], ORCNN [42], PCLDet [56], SFRNet [49], PETDet [26], DRNet [55], and SRCNN [44] by 16.16%, 10.99%, 9.30%, 8.34%, 8.16%, 4.56%, 4.38%, 4.01%, 2.99%, 2.66%, 1.63%, and 1.23%, respectively. These results confirm the overall effectiveness of the PFA-C2FI model on the FAIR1M-2.0 dataset and further demonstrate its strong generalization capability.

F. Computational Cost Analysis

To evaluate the computational cost of the proposed model, quantitative comparisons are conducted in terms of Giga floating point operations (GFLOPs), frames per second (FPS), and the number of parameters against other competing methods. As shown in Table VIII, RoI Trans exhibits the highest computational cost. Apart from this, although the proposed model has relatively more parameters, it achieves comparable GFLOPs and FPS to other methods while obtaining the highest mAP. Overall, the proposed model achieves the best detection accuracy under comparable inference speed.

G. Subjective Evaluation

1) Visualization of Attention Maps and Frequency Responses: To more intuitively illustrate the effectiveness of the PFA-DWT module in enhancing feature representation, the frequency responses of different subbands and the corresponding attention maps are visualized in this section. As shown in Fig. 3, the LL subband mainly captures the primary structural information of the input image, while the LH, HL, and HH subbands characterize texture features in the horizontal, vertical, and diagonal directions, respectively. This is consistent with the objective of employing the PFA-DWT module to enhance directional feature representation for oriented objects. In addition, the attention maps highlight the object regions, indicating that the backbone features refined by attention maps exhibit stronger discriminative capability for object representation.

2) Comparisons With the Baseline Model: This section presents qualitative comparisons between the proposed model and the baseline on the FAIR1M-1.0 and FAIR1M-2.0 testing sets. As shown in Fig. 4(a)–(c), the baseline model incorrectly classifies a Boeing 747, an MB, and a van as OA, OS, and OV, respectively, while the proposed model accurately recognizes all objects. As shown in Fig. 3(d) and (e), the baseline model fails to detect an FF and an intersection, respectively, while the proposed model successfully detects all objects. These qualitative results further demonstrate the superiority of the proposed model over the baseline.

H. Analysis of Failure Cases

As shown in Table VI, the proposed model achieves relatively lower AP on certain subcategories. Therefore, several representative failure cases are selected for further analysis.

1) Misclassification: As shown in Fig. 5(a) and (b), ES is misclassified as warship and FB, respectively. This is mainly attributed to the high visual similarity among certain subcategories under bird's-eye-view imaging conditions.

2) Missed Detection: As shown in Fig. 5(c) and (d), many tractors and TTs are missed. This can be attributed to three main factors. First, the small object size and limited image resolution lead to insufficient feature representation capability. Second, the similarity in color between some objects and the background causes them to be confused with the background. Finally, dense object distribution results in significant overlap among predicted bounding boxes, where the NMS operation

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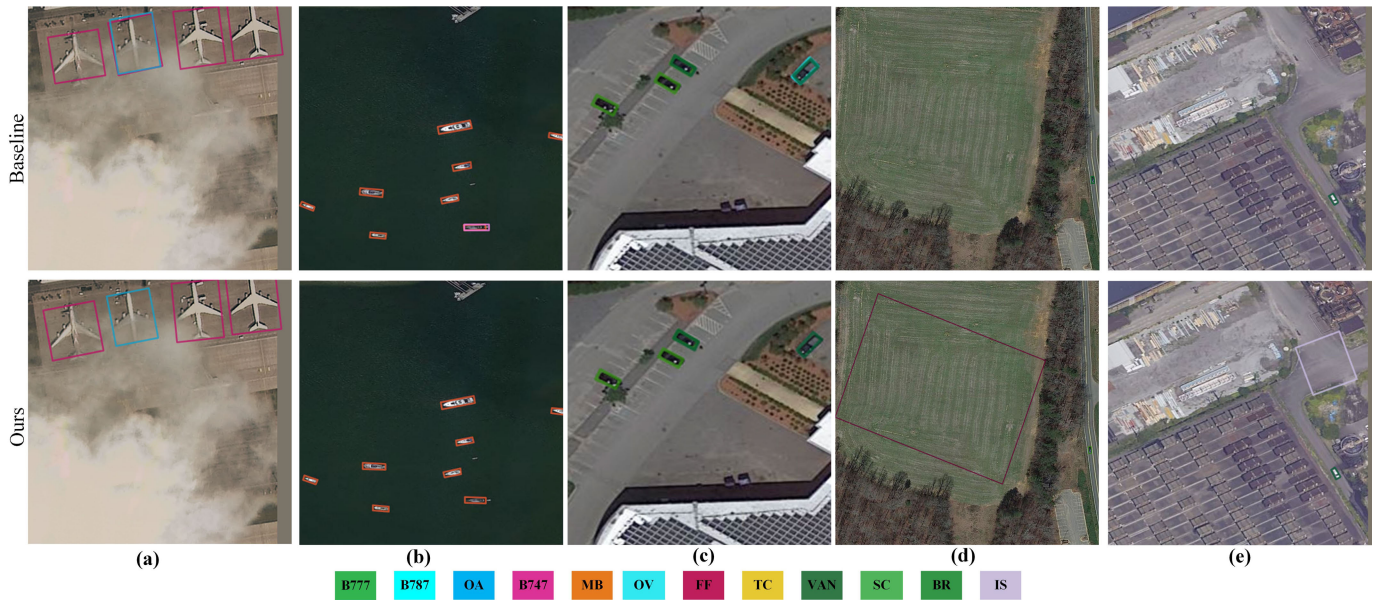


Fig. 4. Subjective comparisons with the baseline model on the FAIR1M-1.0 and FAIR1M-2.0 testing sets. (a) and (e) Selected from the FAIR1M-2.0 testing set, and (b)–(d) selected from the FAIR1M-1.0 testing set.

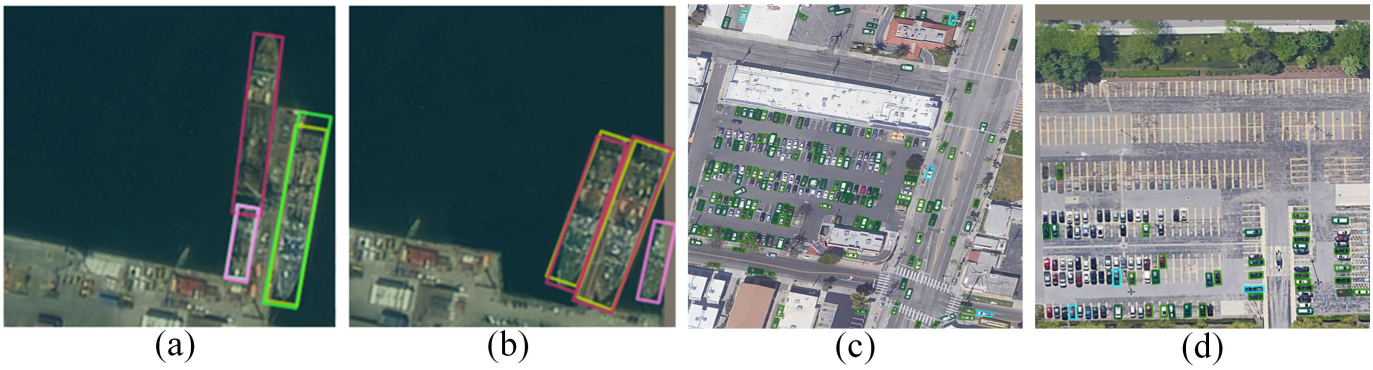


Fig. 5. Visualization of typical failure detection results. (a)–(d) Selected from the FAIR1M-1.0 testing set.

retains only the prediction with the highest confidence score while suppressing other valid detections.

V. Conclusion

This article proposes a novel two-stage FGOD model, namely, PFA-C2FI, which employs a combination of Oriented R-CNN [42] and TF loss [36] as the baseline model, and its main contributions include three aspects. First, a C2FIm-FDE module is proposed and integrated into the backbone network of the baseline model to address the issue of the insufficient discriminability of backbone features. Specifically, the C2FIm-FDE module utilizes generated image-level category labels as guidance and adopts a coarse-to-fine progressive learning strategy to gradually enhance the discriminability of the backbone features for image-level categories. Second, a PFA-DWT module is proposed and incorporated into the backbone network to address the issue of the inadequate representation capability of backbone features for object direction. Specifically, the PFA-DWT module fully leverages the dual localization advantages of DWT in both spatial and frequency domains, enabling it to extract directional information of local objects and achieve accurate pointwise fusion of spatial and frequency-domain features. Finally, a C2FIn head is proposed to further enhance the FGOD model's discriminability for similar subcategories. The C2FIn head guides the FGOD model to focus on the differences among different subcategories within the same coarse-grained category by progressively adding coarse- and fine-grained instance-level supervisions. Ablation studies validate the effectiveness of the PFA-DWT module, the C2FIm-FDE module, and the C2FIn head, as well as any combination thereof. Our PFA-C2FI model achieves mAP of 44.36% and 48.67% on the FAIR1M-1.0 and FAIR1M-2.0 datasets, respectively, and the quantitative comparison with popular FGOD models demonstrates its overall effectiveness.

In future work, our research will focus on two main directions. On the one hand, we will adopt more powerful two-stage FGOD models as a new baseline. On the other hand, we will explore knowledge distillation from large-scale remote sensing foundation models, as well as self-supervised learning with unlabeled data, to further improve the generalization capability of the proposed model.

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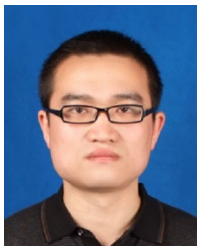
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